

Power

The power supply circuit consists of the following components and connections:

- Input:** A battery (J1, B3B-EH-A) provides +BATT and GND connections.
- Rectification:** A diode (D1, SS2040FL) is connected in series with the +BATT line.
- Regulation:** A voltage regulator (U1, TC1262-3.3VDB) is used to provide a stable +3.3V output. Its VI (input) is connected to the +BATT line, and its V0 (output) is connected to the +3.3V output.
- Decoupling:** Capacitors C2 (22uF) and C3 (0.1uF) are connected to the input of the regulator. Capacitors C4 (0.1uF) and C5 (22uF) are connected to the output of the regulator.
- Output:** The +3.3V output is connected to the BAT_CHECK pin via a resistor network (R1, 62k and R3, 10k).

DF40HC(3.0)-100DS-0.4V

J3

+5V

+3.3V

SPK_L_P

SPK_L_N

S_RXD

BAT_CHECK

LCD_RST

MIC_A

MIC_BIAS_A

SCL

SDA

LCD_SCK

LCD_MOSI

LCD_CS

LCD_BL

TXEN

TXD

RXD

C8

1u

GND

AGND_MIC

[illegible]

The diagram illustrates the pin configuration and power supply connections for the LSF0108PWR component (U3). The component is a yellow rectangle with pins numbered 1 through 20. The connections are as follows:

- Power Supply:**
 - Pin 1: GND
 - Pin 2: VREF_A (+1.8V)
 - Pin 19: VREF_B (+3.3V)
 - Pin 20: EN (+3.3V)
- Signal Connections:**
 - Pin 3: S_RXD
 - Pin 4: S_TXD
 - Pin 5: SD_CLK
 - Pin 6: SD_CMD
 - Pin 7: SD_DAT0
 - Pin 8: SD_DAT1
 - Pin 9: SD_DAT2
 - Pin 10: SD_DAT3
 - Pin 11: B8
 - Pin 12: B7
 - Pin 13: B6
 - Pin 14: B5
 - Pin 15: B4
 - Pin 16: B3
 - Pin 17: B2
 - Pin 18: B1
- Resistors and Capacitors:**
 - Capacitors C10 and C11 (0.1uF) are connected to pins 1 and 19, respectively, to ground.
 - Resistor R7 (180k) is connected to pin 19 to ground.
 - Resistors RN3 and RN4 (1k) are connected to pins 1, 2, 3, and 4, respectively, to ground.
- 3.3V Signal Connections:**
 - Pin 1: S_RXD_3V
 - Pin 2: S_TXD_3V
 - Pin 3: SD_CLK_3V
 - Pin 4: SD_CMD_3V
 - Pin 5: SD_DAT0_3V
 - Pin 6: SD_DAT1_3V
 - Pin 7: SD_DAT2_3V
 - Pin 8: SD_DAT3_3V

The diagram illustrates the wiring for a Micro SD Card Slot (J4, DM3AT) connected to a microcontroller. The connections are as follows:

- Power:**
 - +3.3V:** Connected to pins 1 (DAT2), 2 (DAT3/CD), 3 (CMD), 4 (VDD), 5 (CLK), 6 (VSS), 7 (DAT0), 8 (DAT1), 9 (DET_A), and 10 (DET_B).
 - +1.8V:** Connected to pin 11 (SHIELD).
- Grounding:**
 - GND:** Connected to pins 12 (DAT2), 13 (DAT3/CD), 14 (CMD), 15 (VDD), 16 (CLK), 17 (VSS), 18 (DAT0), 19 (DAT1), 20 (DET_A), and 21 (DET_B).
- Signal:**
 - SD_CD:** Connected to pin 22 (SHIELD).
- Resistor:**
 - R8 (10k):** Connected between the +1.8V supply and the SHIELD pin (pin 11).
- Capacitor:**
 - C9 (0.1u):** Connected between the +3.3V supply and GND.

Extension (I2C)

Diagram showing the I2C extension connection. A 4-pin header J5 (B4B-EH-A) is connected to a circuit. Pin 1 is +5V, Pin 2 is SCL_3V, Pin 3 is SDA_3V, and Pin 4 is GND.

Extension (GPIO)

Diagram showing the GPIO extension connection. A 4-pin header J6 (B4B-EH-A) is connected to a circuit. Pin 1 is +5V, Pin 2 is S_RXD_3V, Pin 3 is S_TXD_3V, and Pin 4 is GND.

Extension (GPIO)

The diagram illustrates the wiring for a GPIO extension using a B4B-EH-A connector (J6). The connector has four pins labeled 1, 2, 3, and 4. The wiring is as follows:

- Pin 1 is connected to +5V.
- Pin 2 is connected to S_RXD_3V.
- Pin 3 is connected to S_TXD_3V.
- Pin 4 is connected to GND.

Speaker

Diagram showing the speaker connection. The SPK_L_P and SPK_L_N signals are connected to a 2-pin header (J8) labeled PinHeader 2.0mm DNP. The pins are numbered 1 and 2.

Microphone

Diagram showing the microphone connection. The MIC_BIAS_A signal is connected to a 2-pin header (J9) labeled PinHeader 2.0mm DNP. The pins are numbered 1 and 2. The MIC_A signal is connected to a 1uF capacitor (C17) which is connected to the AGND_MIC ground. A 2.2k resistor (R11) is connected between the MIC_BIAS_A signal and the AGND_MIC ground. A 0.1uF capacitor (C15) is connected between the MIC_BIAS_A signal and the AGND_MIC ground.

Microphone

MIC_BIAS_A

C15 0.1u

R11 2.2k

AGND_MIC

MIC_A

C17 1u

J9 PinHeader 2.0mm DNP

AGND_MIC